Linear Algebra and Calculus Lecture 4
Multivariable Calculus

Eric Sandin Vidal

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Functions in Several Variables

Definition (12.1.1)**.** A **function** f of n real variables is a rule that assigns a *unique* real number $f(x_1, x_2, \ldots, x_n)$ to each point $(x_1, x_2, \ldots, x_n)$ in some subset $\mathcal{D}(f)$ of $\mathbb{R}^n$ called the **domain** of f. The set of real numbers $f(x_1, x_2, \ldots, x_n)$ obtained from points in the domain is called the **range** of f.

Similarly to when we graph a function in one variable (where we draw the points $(x, f(x))$ in a plane), for functions $f : \mathbb{R}^2 \to \mathbb{R}$, we will plot the graph of f by drawing the points $(x, y, f(x, y))$ on a 3-dimensional space.

Example. Consider $f(x, y) = \sqrt{9 - x^2 - y^2}$. The domain is made up of the points for which the expression under the square root is non-negative, that is $9 - x^2 - y^2 \geq 0$, or in other words, the disk __________. To plot this, we can take $z = f(x, y)$ so $z = \sqrt{9 - x^2 - y^2}$ and hence ______________. This is a sphere of radius 3 centered at the origin. However, the graph consists only of the positive square root, and therefore where $z \geq 0$.

Example. We can also have functions that go from $\mathbb{R}^m$ to $\mathbb{R}^n$, for instance $f : \mathbb{R}^3 \to \mathbb{R}^2$ with $f(x, y, z) =$ ______________.

Limits

In this section we consider functions $f : \mathbb{R}^n \to \mathbb{R}$, so when we mention $\mathbf{x} \in \mathbb{R}^n$, it will mean that $\mathbf{x} = (x_1, x_2, \ldots, x_n)$. The following definitions are *non-formal*. The actual definitions of limits in several variables require more nuance and prior results in topology in $\mathbb{R}^n$.

Definition (12.2.2)**.** We say that $\lim_{\mathbf{x} \to \mathbf{x}^*} f(\mathbf{x}) = L$, provided that

- every neighbourhood of $\mathbf{x}^*$ contains ______________________ different from $\mathbf{x}^*$
- $f(\mathbf{x})$ gets as close as we want to L when _________________________.

The same properties that held for limits in one variable, also hold for several variables.

Example. Here are some examples:

- $\lim_{(x,y) \to (2,3)} (2x - y^2) = 4 - 9 = -5$

- $\lim_{(x,y)\to(a,b)} x^2y =$ ___

- $\lim_{(x,y)\to(\frac{\pi}{3},2)} y\sin\left(\frac{x}{y}\right) = 2\sin\left(\frac{\pi}{6}\right) = 1.$

Remark. In this course we will not deal with limits to points outside of the domain such as $\lim_{(x,y)\to(0,0)} \frac{2xy}{x^2+y^2}$, since they require much more attention to detail as we can approach such points from many different directions.

Definition (12.2.3)**.** The function $f : \mathbb{R}^n \to \mathbb{R}$ is continuous at $\mathbf{x}^*$ if

$$\lim_{\mathbf{x}\to\mathbf{x}^*} f(\mathbf{x}) = f(\mathbf{x}^*).$$

The properties for continuous functions that hold for single variable functions also hold for several variable functions.

Partial Derivatives

If we talk about the slope of the tangent line of a function at a certain point, we can actually wonder, in which direction? Thus, the definition of derivative in the multivariable case is adaptable to that idea.

Remark. For simplicity purposes, we will give the following definitions for functions in $\mathbb{R}^2$, but they extend naturally to functions in $\mathbb{R}^n$.

Definition. The **first partial derivatives** of the function $f(x, y)$ ____________________ ________________ are the functions $f_1(x, y)$ and $f_2(x, y)$ given by

$$\frac{\partial}{\partial x} f(x,y) = f_1(x,y) = \lim_{h\to 0} \frac{f(x+h,y) - f(x,y)}{h},$$
$$\frac{\partial}{\partial y} f(x,y) = f_2(x,y) = \underline{\qquad\qquad\qquad\qquad}.$$

Example. Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ if $f(x, y) = x^3y^2 + x^4y + y^4$.

$$\frac{\partial f}{\partial x} = 3x^2y^2 + 4x^3y,$$
$$\frac{\partial f}{\partial y} = \underline{\qquad\qquad\qquad}.$$

Exercise. Find $f_1(0, \pi)$ if $f(x, y) = e^{xy} \cos(x + y)$.

Notice that the **Chain Rule** applies all the same to partial derivatives.

Definition (12.5)**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ and $g_i : \mathbb{R} \to \mathbb{R}$ be functions such that x maps to $f(g_1(x), g_2(x), \ldots, g_n(x))$. If all the partial derivatives of f exist and are continuous and g_i is differentiable for all $i = 1, \ldots, n$, then

$$\frac{df}{dx} = \frac{d}{dx} f(g_1(x), g_2(x), \ldots, g_n(x)) = \underline{\qquad\qquad}.$$

Example. If f is a function of x and y with continuous first partial derivatives, and if x and y are differentiable functions of t, then

$$\frac{df}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}.$$

Example. If $f(x,y) = 4x^2 + 3y^2$, where $x = x(t) = \sin t$ and $y = y(t) = \cos t$, then

$$\frac{d}{dt} f(x(t), y(t)) = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} = 8x\cos t - 6y\sin t,$$

now plug back the expressions of $x(t)$ and $y(t)$:

$$\frac{d}{dt} f(x(t), y(t)) = \underline{\qquad\qquad}.$$

Notice also that we can do higher order partial derivatives by differentiating with respect one variable first, and another later.

Definition (12.4)**.** Higher order derivatives are defined as follows;

$$\begin{aligned}
\frac{\partial^2}{\partial x^2} f(x,y) &= \frac{\partial}{\partial x}\frac{\partial}{\partial x} f(x,y) = f_{11}(x,y) = f_{xx}(x,y),\\
\frac{\partial^2}{\partial y^2} f(x,y) &= \frac{\partial}{\partial y}\frac{\partial}{\partial y} f(x,y) = f_{22}(x,y) = f_{yy}(x,y),\\
\frac{\partial^2}{\partial x \partial y} f(x,y) &= \frac{\partial}{\partial x}\frac{\partial}{\partial y} f(x,y) = f_{21}(x,y) = f_{yx}(x,y),\\
\frac{\partial^2}{\partial y \partial x} f(x,y) &= \frac{\partial}{\partial y}\frac{\partial}{\partial x} f(x,y) = f_{12}(x,y) = f_{xy}(x,y).
\end{aligned}$$

Remark. Differentiating twice with respect to different variables is **in general** a commutative operation, so $\dfrac{\partial^2}{\partial x \partial y} f(x,y) = \dfrac{\partial^2}{\partial y \partial x} f(x,y)$.

A vector $\mathbf{f} = (f_1, f_2, \ldots, f_m)$ of m functions, each depending on n variables $(x_1, x_2, \ldots, x_n)$, defines a *map* (i.e., a function) $f : \mathbb{R}^n \to \mathbb{R}^m$, so if $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ then

$$\mathbf{f}(\mathbf{x}) = \begin{cases} f_1(x_1, \ldots, x_n) \\ f_2(x_1, \ldots, x_n) \\ \vdots \\ f_m(x_1, \ldots, x_n). \end{cases}$$

Information about the rate of change of $\mathbf{f}(\mathbf{x})$ with respect to $\mathbf{x}$ is contained in the various partial derivatives, which is conveniently organized into an $m \times n$ matrix, $\mathrm{D}\mathbf{f}(\mathbf{x})$, called the ________________ of $\mathbf{f}$:

$$\mathrm{D}\mathbf{f}(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}.$$

If the partial derivatives in the Jacobian matrix are continuous, we say that $\mathbf{f}$ is **differentiable** at $\mathbf{x}$. Notice that the Chain Rule also applies: __________________________.

Example. Let $\mathrm{f} : \mathbb{R}^2 \to \mathbb{R}^3$ given by $\mathrm{f}(x, y) = (xe^y + \cos(\pi y), x^2, x - e^y)$. Find the Jacobian $D\mathrm{f}(1, 0)$.

$$D\mathrm{f}(1,0) = \begin{bmatrix} e^y & \underline{\qquad\qquad\qquad} \\ \_\_ & 0 \\ 1 & -e^y \end{bmatrix}\Bigg|_{(1,0)} = \begin{bmatrix} 1 & 1 \\ 2 & 0 \\ 1 & -1 \end{bmatrix}.$$

Exercise. Let $\mathrm{f} : \mathbb{R}^2 \to \mathbb{R}^2$ given by $\mathrm{f}(x, y) = (x^2 y, 5x + \sin y)$. Find the Jacobian $D\mathrm{f}(x, y)$.

Endgame Content: Gradients

Let us go now with a definition of a concept that requires some Linear Algebra knowledge but that is of great importance in many areas of Computer Science and Machine Learning, especially in optimization problems.

Definition (12.7.6)**.** At any point $p = (p_1, \ldots, p_n)$ where the first partial derivatives of a function $f : \mathbb{R}^n \to \mathbb{R}$ exist, we define the ________________ $\nabla f(p) = \operatorname{grad} f(p)$ by

$$\nabla f(p) = \operatorname{grad} f(p) = \begin{bmatrix} \dfrac{\partial f}{\partial x_1}(p_1) \\ \vdots \\ \dfrac{\partial f}{\partial x_n}(p_n) \end{bmatrix}.$$

Before, we define first partial derivatives as the rate of change of a function in the x or y directions (or the directions of any other variable). But what if we want to find the rate of change in any other direction given by a vector (or direction) $\mathbf{u}$? We will not give an explicit definition of directional derivatives, just like we have not given full definitions of the notion of differentiability (not partial differentiability), but we will give a very useful result regarding gradients.

Theorem (12.7.7)**.** *If f is differentiable at $p = (p_1, \ldots, p_n)$ and $\mathbf{u} = (u_1, \ldots, u_n)$ is a unit vector, then the directional derivative of f at p in the direction of $\mathbf{u}$ is given by*

$$D_{\mathbf{u}} f(p) = \mathbf{u} \cdot \nabla f(p) = \sum_{i=1}^{n} u_i \frac{\partial f}{\partial x_i}(p_i) = \underline{\qquad\qquad\qquad\qquad\qquad},$$

that is, the dot product of $\mathbf{u}$ and the gradient of f at p.

These results are used in processes and algorithms such as **gradient descent**, in which the fact that a function f increases (or decreases) most rapidly at a point p in the direction of its gradient (or the negative of the gradient) is used to find most optimal paths.